30	**Answer: C**
	Let B be the background count rate. The actual initial count rate, $C_0 = 200 - B$ After four half-lives have elapsed, the total count rate, $0.25\ (200) = 50$ The actual count rate, $C = 50 - B$ Using $\frac{C}{C_0} = \left(\frac{1}{2}\right)^n$ and that $n = 4$,

	$\frac{50-B}{200-B}=\left(\frac{1}{2}\right)^4=\frac{1}{16}$ $\therefore 800-16B=200-B$ $15B=600$ $B=40 \text{ counts min}^{-1}$